Software Requirements Specification (SRS)

Project: Online Exam Portal

Authors: Krishna H Zalavadiya, DEERAJ S, Kondapuram Charith Sai Reddy, Korada Sannath Kumar

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Status:

Revision history

Version	Date	Author	Change summary	Approval
1.0	06-09-2026	Student	Initial SRS drafted for Online Exam Portal (SecureExam-AI) based on project repository reference	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	Sheela Devi		

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Online Exam Portal, developed as the. It defines the functional and non-functional requirements, interfaces, and verification criteria for a web-based platform that conducts secure online examinations with AI-assisted proctoring, automated evaluation, and administrative analytics.

1.2 Scope

Covers student registration and authentication, exam creation and management by administrators, online exam attempt with a live timer and auto-submission, AI-based proctoring (face monitoring, tab-switch and fullscreen-exit detection), automatic evaluation, result publication, admin analytics dashboards, and Excel-based report export. Excludes live human-invigilator video calling, payment/billing modules, and integration with external university ERP systems (out of scope for the current version).

1.3 Audience

Developers, QA Engineers, Project Guide/Evaluator, System Administrators, and end users (Students and Admins).

1.4 Definitions

List of acronyms: SRS, UI, API, REST, JWT, ODM, DB, MCQ, AI, CRUD, RBAC, WCAG, TLS.

2. Overall description

2.1 Product perspective

The Online Exam Portal is a self-contained MERN-stack (MongoDB, Express.js, React, Node.js) web application. It consists of a React + Vite single-page frontend, an Express.js REST API backend, a MongoDB data store, a JWT-based authentication layer, and a browser-based AI proctoring module that uses the device camera and browser events to monitor student behaviour during an exam attempt.

2.2 Major product functions (detailed)

- Student registration and secure login
- Admin creation, editing, and deletion of exams and questions
- Student exam attempt in a fullscreen, timed interface with automatic submission
- AI-based proctoring: face monitoring, tab-switch detection, fullscreen-exit detection, and violation counting
- Automatic evaluation and result generation
- Result history and profile management for students
- Admin analytics dashboard (totals, performance, score distribution)
- Excel report export of student results

2.3 User roles and characteristics (expanded)

- Student: basic computer literacy, attempts exams within a fixed time window and expects a responsive, distraction-free interface.
- Admin: manages the exam lifecycle (create/edit/delete exams and questions), reviews attempts, results, and analytics, and exports reports.

2.4 Operating environment

Modern evergreen web browsers (Chrome, Edge, Firefox) with camera access permission, on desktop or laptop devices with a stable internet connection. Backend deployed on a Node.js runtime with a MongoDB database (self-hosted or Atlas).

2.5 Constraints

Passwords must be hashed (bcryptjs); sessions authenticated via JWT; exam attempt requires fullscreen and camera permission; question banks and answer keys must not be exposed on student-facing API responses; report export limited to authorised Admin roles.

3. External interface requirements

3.1 User interfaces

Primary UI: a responsive React + Tailwind CSS web interface with separate Student and Admin dashboards, a distraction-minimised fullscreen exam-taking screen, and a live countdown timer.

3.2 Hardware interfaces

- Device webcam (for AI face-monitoring proctoring)
- Keyboard and mouse / trackpad for standard interaction
- Standard display for exam UI and analytics charts

3.3 Software interfaces

- Backend REST API (Express.js) over JSON for authentication, exams, questions, results, admin stats, and export
- MongoDB (via Mongoose ODM) for persistent storage of users, exams, questions, attempts, and results
- ExcelJS library for generating downloadable Excel result reports
- Recharts library for rendering analytics charts on the Admin dashboard

3.4 Communications

- HTTPS/TLS for all client-server communication in production
- JWT bearer tokens attached to authenticated API requests; Axios used on the frontend for API calls

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow OEP-F-###.

4.1 Authentication & User Management

Description: Register and authenticate Students and Admins, and protect all subsequent API access with role-aware JWT sessions.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-001	The system shall allow a new student to register with name, email, and password.	Functional	High	Student	AC-OEP-F-001: Valid registration creates an account and redirects to login. Test: TC-Auth-01	Email must be unique
OEP-F-002	The system shall authenticate users (Student/Admin) via email and password and issue a signed JWT on success.	Functional	High	Security	AC-OEP-F-002: Correct credentials return a valid JWT; incorrect credentials are rejected. Test: TC-Auth-02	Depends on bcrypt hash verification
OEP-F-003	The system shall hash and store passwords using bcryptjs; plaintext passwords shall never be persisted.	Functional	High	PCI/Security	AC-OEP-F-003: Database inspection shows only hashed values. Test: TC-Auth-03	Audit review required
OEP-F-004	The system shall restrict Admin-only routes (exam/question management, analytics, export) using role-based access control.	Functional	High	Security	AC-OEP-F-004: Student token receives 403 on Admin routes. Test: TC-Auth-04	Requires role claim in JWT
OEP-F-005	The system shall allow a logged-in student to view	Functional	Medium	Student	AC-OEP-F-005: Profile changes persist and are	

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	and update their profile information.				reflected on next login. Test: TC-Auth-05	

4.2 Exam Creation & Management (Admin)

Description: Allow Admins to manage the full exam lifecycle, including exams and their question sets.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OEP-F-010	The system shall allow an Admin to create a new exam with title, duration, and schedule.	Functional	High	Admin	AC-OEP-F-010: Created exam appears in the exam list. Test: TC-Exam-01	
OEP-F-011	The system shall allow an Admin to edit an existing exam's details.	Functional	High	Admin	AC-OEP-F-011: Edits are saved and reflected immediately. Test: TC-Exam-02	
OEP-F-012	The system shall allow an Admin to delete an exam, provided no active attempts exist.	Functional	Medium	Admin	AC-OEP-F-012: Deleted exam is removed from listings. Test: TC-Exam-03	Referential integrity with attempts/results
OEP-F-013	The system shall allow an Admin to add one or more questions (with options and correct answer) to an exam.	Functional	High	Admin	AC-OEP-F-013: Added question is stored and linked to the exam. Test: TC-Q-01	
OEP-F-014	The system shall allow an Admin to edit or delete existing questions.	Functional	Medium	Admin	AC-OEP-F-014: Change is reflected in the question bank. Test: TC-Q-02	
OEP-F-015	The system shall list all registered students and their exam attempts for Admin review.	Functional	Medium	Admin	AC-OEP-F-015: Student and attempt lists match database records. Test: TC-Admin-01	

4.3 Exam Attempt (Student)

Requirements: The student shall view available exams, start an attempt in fullscreen mode, answer questions within a live countdown timer, and have the attempt auto-submitted when time expires or on manual submit. IDs: OEP-F-020..OEP-F-025.

4.4 AI Proctoring

Requirements: During an active attempt, the system shall enable the camera for face monitoring, detect tab-switching and exits from fullscreen mode, increment a violation counter for each infraction, and automatically submit the exam once a configured violation threshold is reached. IDs OEP-F-030..OEP-F-034.

4.5 Result Evaluation, Analytics & Export

Requirements: The system shall automatically evaluate objective answers on submission, publish results and history to the student, aggregate exam/student/attempt statistics and score distribution on the Admin

analytics dashboard, and allow the Admin to export results (student name, email, exam, score, percentage, submission date) to an Excel file via ExcelJS. IDs OEP-F-040..OEP-F-043.

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs OEP-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
OEP-NF-001	Exam page load and question navigation shall respond within 2 seconds for 90% of requests under normal load.	Performance	High	90th percentile ≤ 2s in load test. Test: TC-Perf-01
OEP-NF-002	The platform shall provide 99.5% availability during scheduled exam windows.	Reliability	High	Uptime monitoring shows ≥99.5% during exam slots. Test: Ops reports
OEP-NF-003	All stored passwords and JWT secrets shall comply with secure-storage practice; no sensitive data in plaintext.	Security/Compliance	High	Security checklist / code review pass. Test: TC-Sec-01
OEP-NF-004	The system shall retain exam attempt and violation logs for at least 1 academic year for audit purposes.	Audit/Data Retention	Medium	Log retention verified in DB backup policy. Test: TC-OPS-01
OEP-NF-005	The UI shall be usable on common screen sizes (1366×768 and above) and follow WCAG 2.1 AA colour-contrast guidance where applicable.	Usability/Accessibility	Medium	Accessibility spot-check pass. Test: TC-UX-01

5.1. Security

5.1.1 Security Objectives

1. Prevent exam malpractice and impersonation through continuous AI-based proctoring and violation-based auto-submission.

2. Protect the confidentiality and integrity of student credentials, question banks/answer keys, and results through encryption, hashing, and role-based access control.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
PRJ-SR-001	TLS 1.2+ mandatory for all network connections between client and server in production.	Security	High	TLS enforced check. Test: TC-Sec-02
PRJ-SR-002	The system shall hash all passwords with bcryptjs (salted) and never log or store plaintext passwords.	Security	High	Code review + DB inspection. Test: TC-Sec-03
PRJ-SR-003	The system shall issue short-lived, signed JWTs carrying a role claim and reject expired or tampered tokens.	Security	High	Expired/altered token rejected. Test: TC-Sec-04
PRJ-SR-004	Question bank correct answers shall never be included in any student-facing API response before submission.	Security	High	API response review shows no answer keys pre-submission. Test: TC-Sec-05
PRJ-SR-005	AI proctoring violation events shall be recorded in an append-only, timestamped audit log per attempt.	Security	Medium	Audit log entries verified per test attempt. Test: TC-Sec-06

6. Quality attributes & Acceptance tests

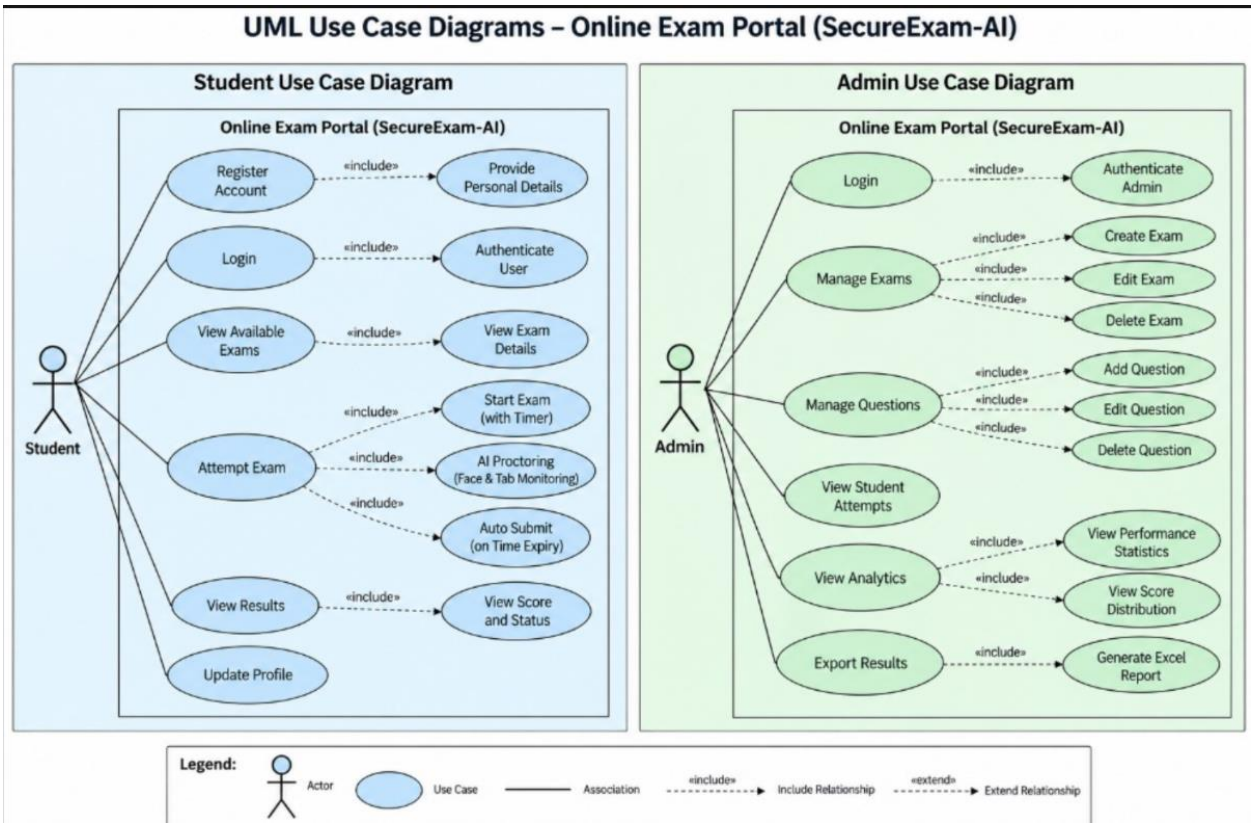
- Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR failures, and the RTM shows all test cases passed.

- Acceptance test suites: Authentication, Exam Management, Exam Attempt, AI Proctoring, Result & Analytics, Security, and Usability tests.

7. System models and diagrams

7.1 UML Use-Case diagram

At least 2 UML use-case diagrams to be created: (1) Student use cases — Register, Login, View Exams, Attempt Exam, View Results; (2) Admin use cases — Login, Create/Edit/Delete Exam, Manage Questions, View Analytics, Export Results.



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
OEP-F-002	Login & JWT issue	4.1 / DS-Auth-01	AuthModule	TC-Auth-02	N	
OEP-F-010	Create exam	4.2 / DS-Exam-01	ExamModule	TC-Exam-01	N	
OEP-F-013	Add question	4.2 / DS-Exam-02	QuestionModule	TC-Q-01	N	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
OEP-F-030	AI proctoring / violation count	4.4 / DS-Proctor-01	ProctorModule	TC-Proct-01	N	
OEP-F-040	Auto evaluation & result	4.5 / DS-Result-01	ResultModule	TC-Result-01	N	
OEP-NF-001	Response time target	5 / DS-Perf-01	WebUI / API	TC-Perf-01	N	
PRJ-SR-002	Password hashing	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-03	N	